Literature-Implied Partial Answer to Yun's L_r-to-L_s Question

FA/Banach run packet

June 16, 2026

Status

Result type: literature implied answer (partial subcase).

Source question: Longyun Ding, *Borel reducibility and Hölder(α) embeddability between Banach spaces*, arXiv:0912.1912, Section 7.

Supporting theorem: Assaf Naor, *Discrete Riesz transforms and sharp metric X_p inequalities*, arXiv:1601.03332.

Source Question

In the final remarks of arXiv:0912.1912, after noting that little is known for $r \geq 2$, Yun asks:

$$\text{For } r, s \geq 2,\ r \leq s, \text{ does } L_r \text{ finitely Lipschitz embed into } L_s?$$

The same displayed question also asks whether this would imply, or accompany,

$$E(L_r, p) \leq_B E(L_s, p).$$

Identification

Yun proves earlier in the same paper that, for Banach spaces X, U and $\alpha > 0$, X finitely Hölder(α)-embeds into U if and only if X Hölder(α)-embeds into an ultrapower $(U)_{\mathcal{U}}$.

Naor proves sharp metric X_p inequalities. One consequence, stated in the abstract of arXiv:1601.03332, is that for $2 < q < p < \infty$, the maximal $\theta \in (0, 1]$ for which L_q admits a bi-θ-Hölder embedding into L_p is exactly q/p. In particular, no bi-Lipschitz embedding $L_q \to L_p$ exists when $2 < q < p < \infty$.

Formal Consequence

Theorem. *For $2 < r < s < \infty$, L_r does not finitely Lipschitz embed into L_s. Consequently, the finite-Lipschitz part of Yun's Section 7 question has a negative answer in the strict super-Hilbertian range.*

Proof. Suppose that L_r finitely Lipschitz embeds into L_s for some $2 < r < s < \infty$. By Yun's ultraproduct characterization with $\alpha = 1$, there is a bi-Lipschitz embedding

$$L_r \longrightarrow (L_s)_{\mathcal{U}}$$

for a free ultrafilter $\mathcal{U}$.

An ultrapower of an L_s-space is again an abstract L_s-space. Naor's sharp X_p inequality applies to L_s-spaces and bounds the possible bi-Hölder exponent of an embedding of L_r into such a target by r/s. Since $r/s < 1$, exponent 1 is impossible. This contradicts the bi-Lipschitz embedding above. □

Scope and Remaining Questions

This is not a full answer to Yun's final questions. The endpoint $r = s$ is trivial, and the Hilbertian source case $r = 2$ is outside the negative argument. The packet does not settle the Borel reducibility assertion $E(L_r, p) \leq_B E(L_s, p)$, nor Yun's separate question about finite Hölder(p/q) self-embeddings of L_r and $E(L_r, p) \leq_B E(L_r, q)$.

The supporting paper does not explicitly cite Yun's question; this is an agent-identified implication, so the packet belongs under `literature_implied_answers/`, not `literature_already_answered/`.

Verification

Stored files:

`source_paper.pdf`, `supporting_paper_1601.03332.pdf`.

Cheap-index search before packaging found no exact prior packet for arXiv:0912.1912. Local corpus and web searches for the exact $E(L_r, p)$, $E(\ell_r, p)$, finite Lipschitz, and Hölder-embeddability phrases found no existing run packet for this target. The decisive later theorem was identified through the metric X_p / snowflake literature.

References

[1] L. Yun, *Borel reducibility and Hölder*(α) *embeddability between Banach spaces*, arXiv:0912.1912. https://arxiv.org/abs/0912.1912

[2] A. Naor, *Discrete Riesz transforms and sharp metric* X_p *inequalities*, Ann. of Math. (2) 184 (2016), no. 3, 991–1016; arXiv:1601.03332. https://arxiv.org/abs/1601.03332